Nucleic Acids
Lecturer 6

Presented By

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1- RNA Structure

- RNA and DNA are both nucleic acids, but differ in main ways.
- **Comparison with DNA:**

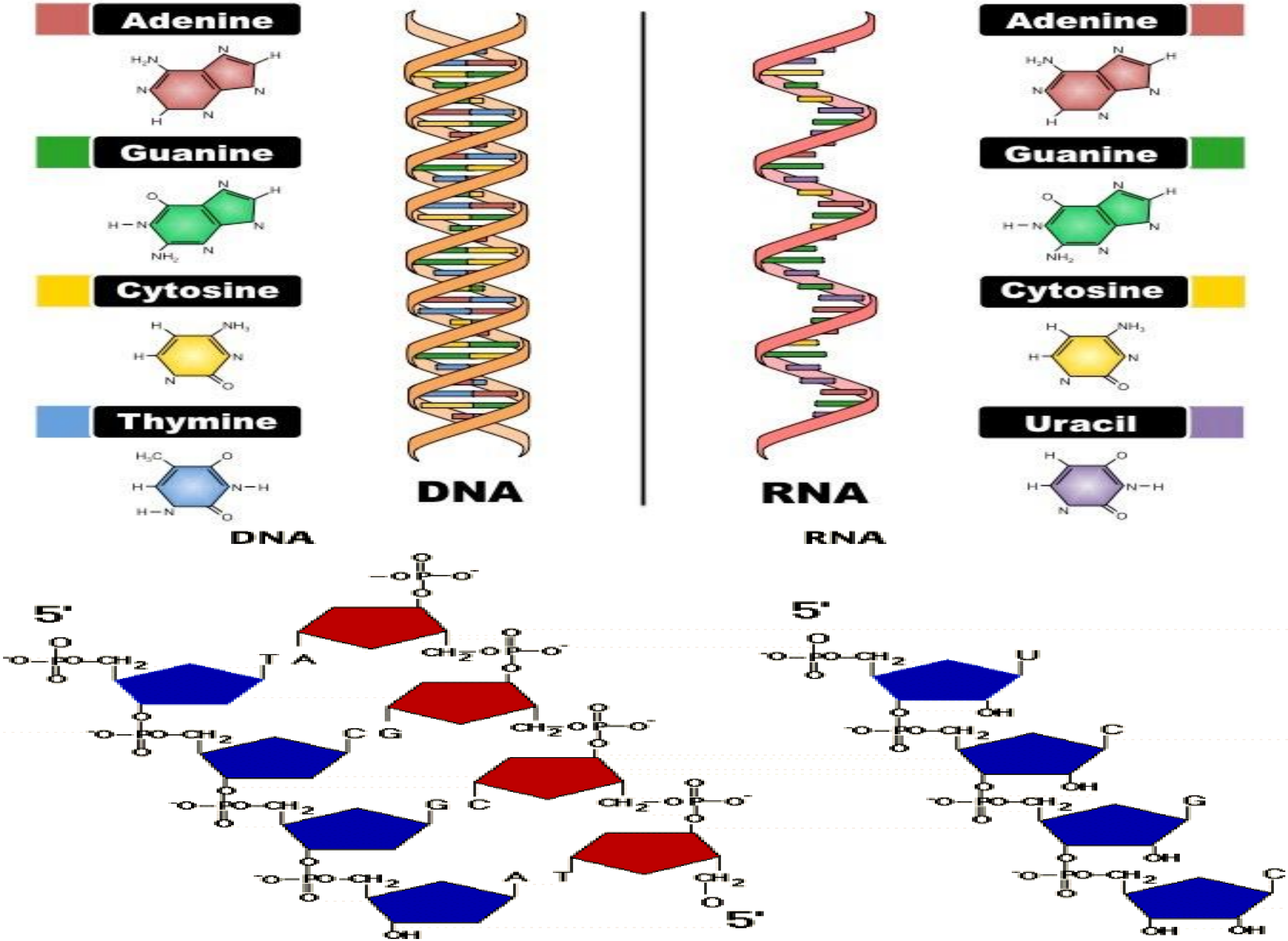
1. Unlike double-stranded DNA, RNA is a single-stranded molecule in many of its biological roles. RNA can also form secondary structures in which a single RNA molecule folds over and forms hairpin loops, stabilized by hydrogen bonds between complementary bases.
2. RNA has a much shorter chain of nucleotides.
3. While DNA contains deoxyribose, RNA contains ribose .These hydroxyl groups make RNA less stable than DNA because it is more prone to hydrolysis.

1- RNA Structure

4. The complementary base to adenine is not thymine, as it is in DNA, but rather uracil, which is an unmethylated form of thymine.

5. Since the RNA molecule is a single strand complementary to only one of the two strands of a gene, its guanine content does not necessarily equal its cytosine content, nor does its adenine content necessarily equal its uracil content.

6. RNA can be hydrolyzed by alkali to 2′,3′ cyclic diesters of the mononucleotides, compounds that cannot be formed from alkali-treated DNA because of the absence of a 2′-hydroxyl group. The alkali lability of RNA is useful both diagnostically and analytically.



Adenine
Guanine
Cytosine
Thymine
DNA
RNA
Adenine
Guanine
Cytosine
Uracil
DNA
RNA
5'
T A
C G
G C
A T
5'
5'
U
C
G
C

Chapter Three: RNA

- 1- RNA Structure

- 2- RNA Types

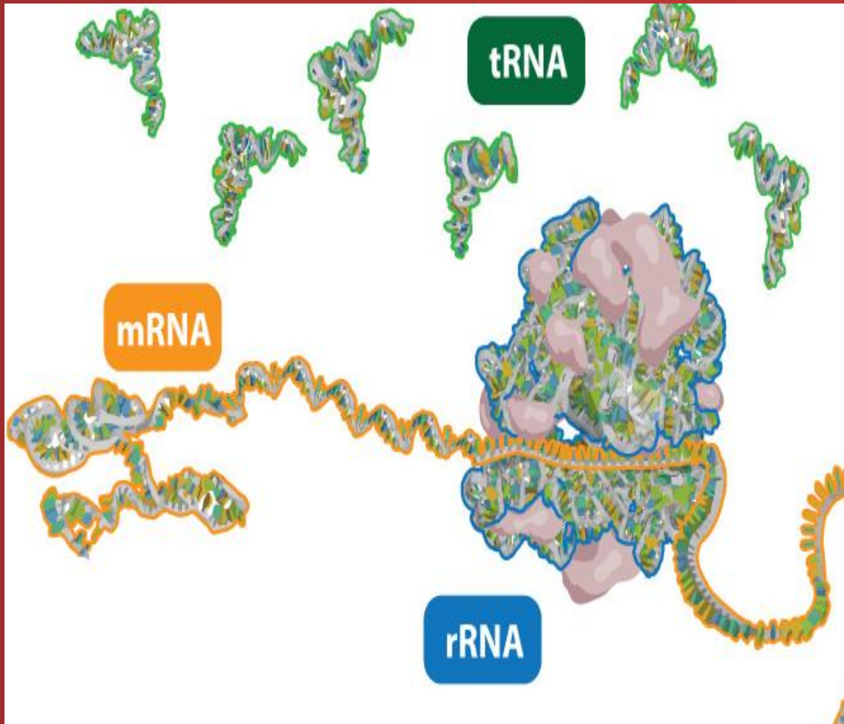
2- RNA Types

- Nearly all the Species of RNA are involved in some aspect of protein synthesis. Each differs from the others by size, function, and general stability.
- In all prokaryotic and eukaryotic organisms, three main classes of RNA molecules exist:

a) Messenger RNA (**mRNA**).

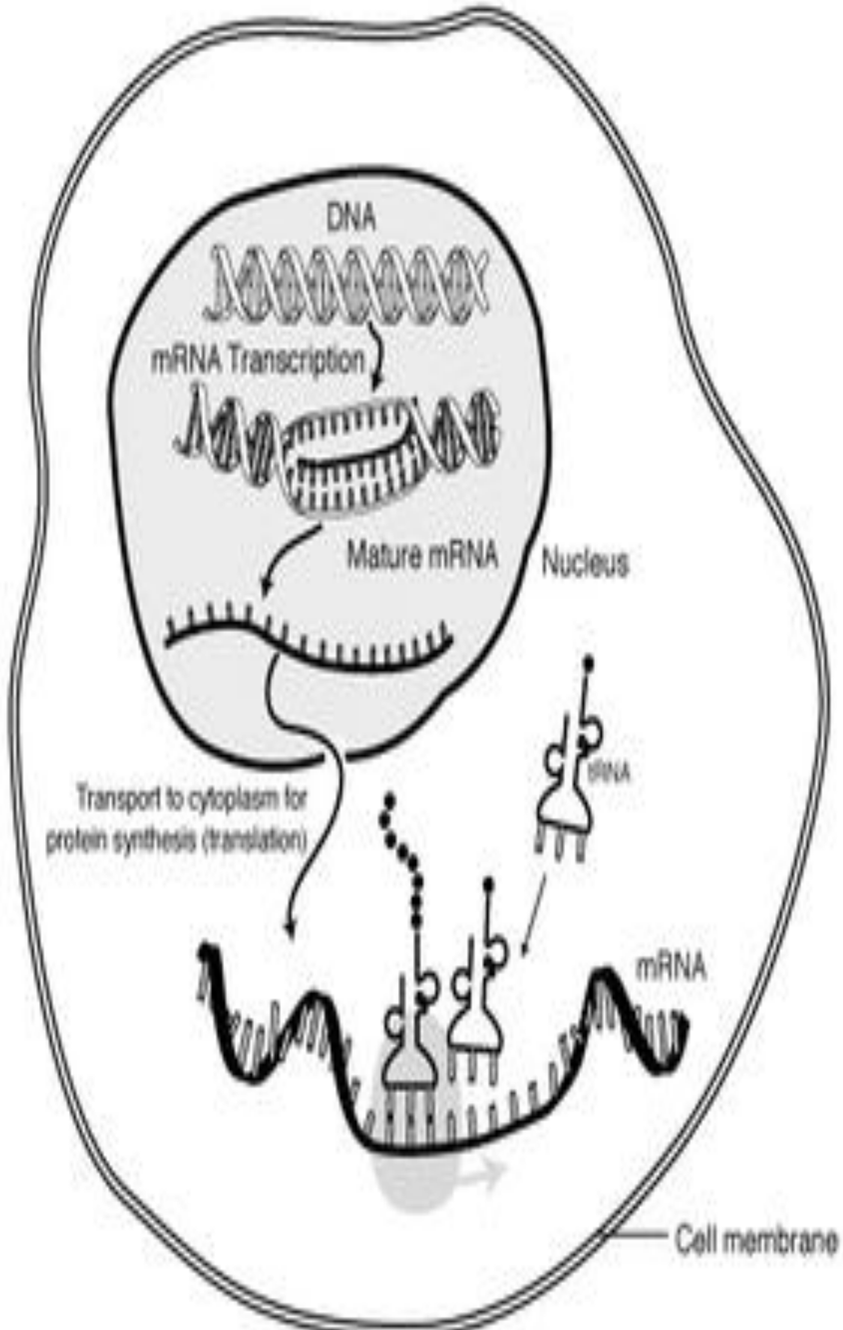
b) Transfer RNA (**tRNA**).

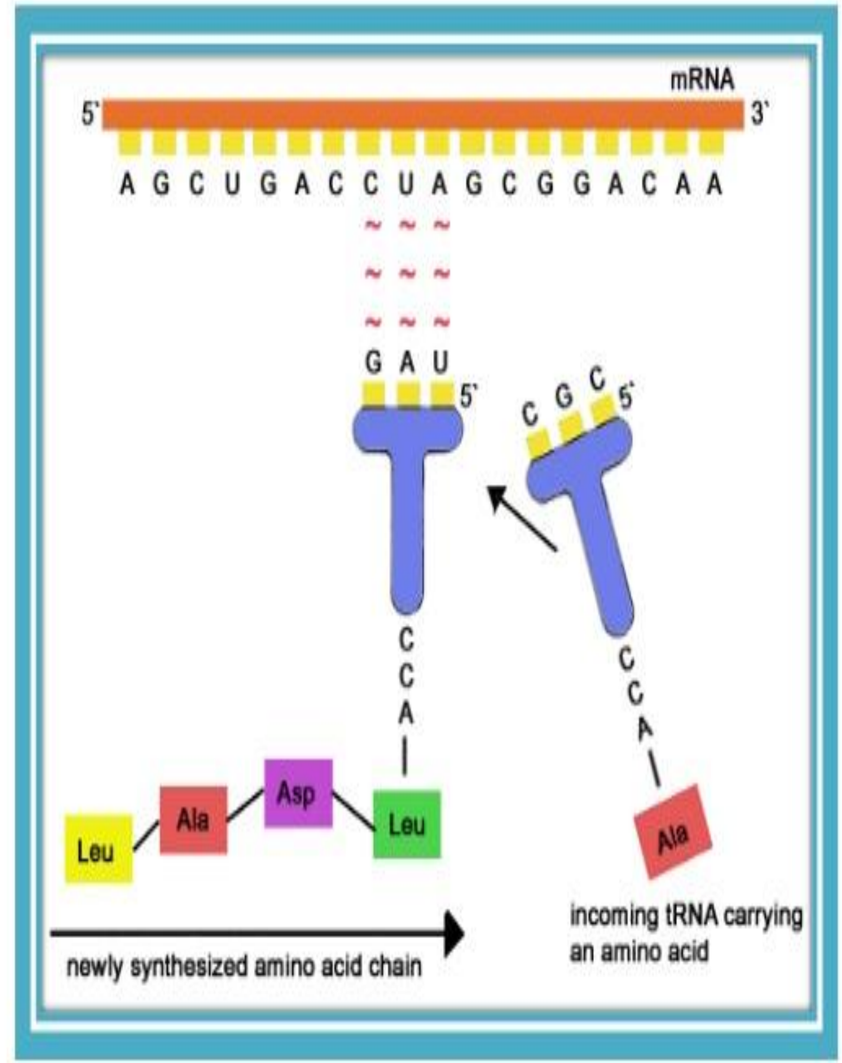
c) Ribosomal RNA (**rRNA**).



a- Messenger RNA (mRNA):

- Is the RNA that carries information from DNA to the ribosome, the sites of protein synthesis (translation) in the cell.
- The coding sequence of the mRNA determines the amino acid sequence in the protein that is produced. It is coded so that every three nucleotides (a codon) correspond to one amino acid.
- In eukaryotic cells, once precursor mRNA (**pre-mRNA**) has been transcribed from DNA, it is processed to **mature mRNA**
- The mRNA is then exported from the nucleus to the cytoplasm, where it is bound to ribosomes and translated into its corresponding protein form with the help of tRNA..
- After a certain amount of time the message degrades into its component nucleotides with the assistance of ribonucleases.

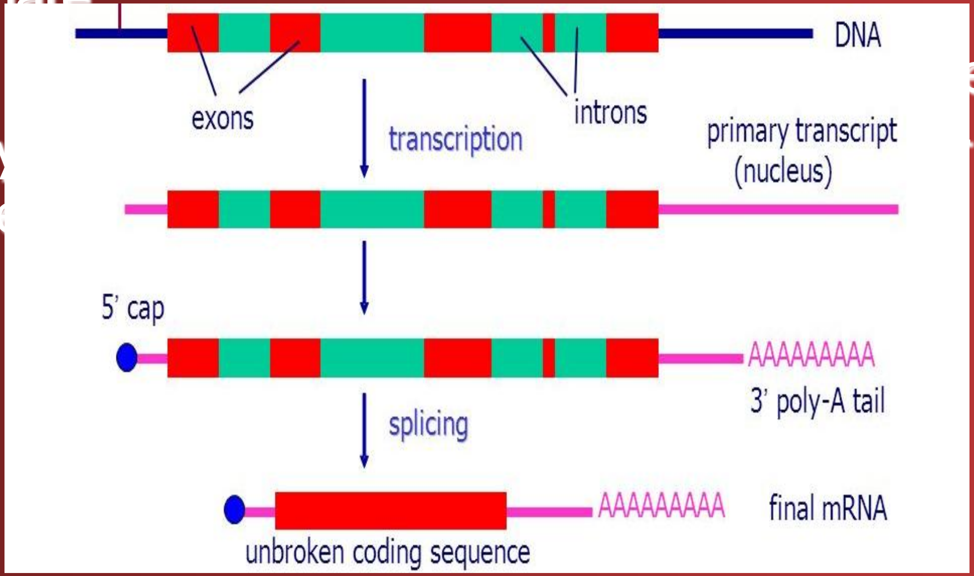




MESSENGER RNA FURNISHED WITH GENETIC CODE

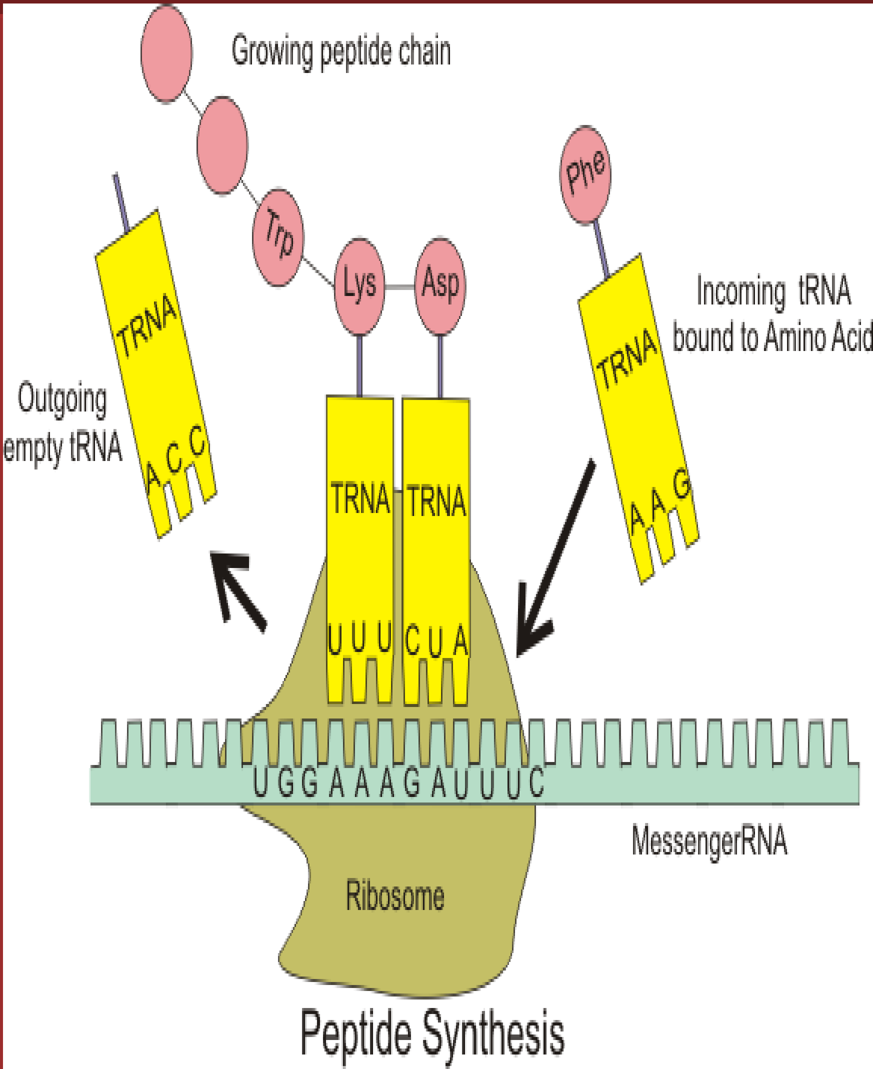
Processed mRNA

- **Mature mRNA** containing:
- The 5′ terminal of mRNA is capped by a 7-methylguanosine triphosphate
- Removal [obscured] e pre-mRNA.
- The 3′-hy[obscured] of adenylat[obscured]



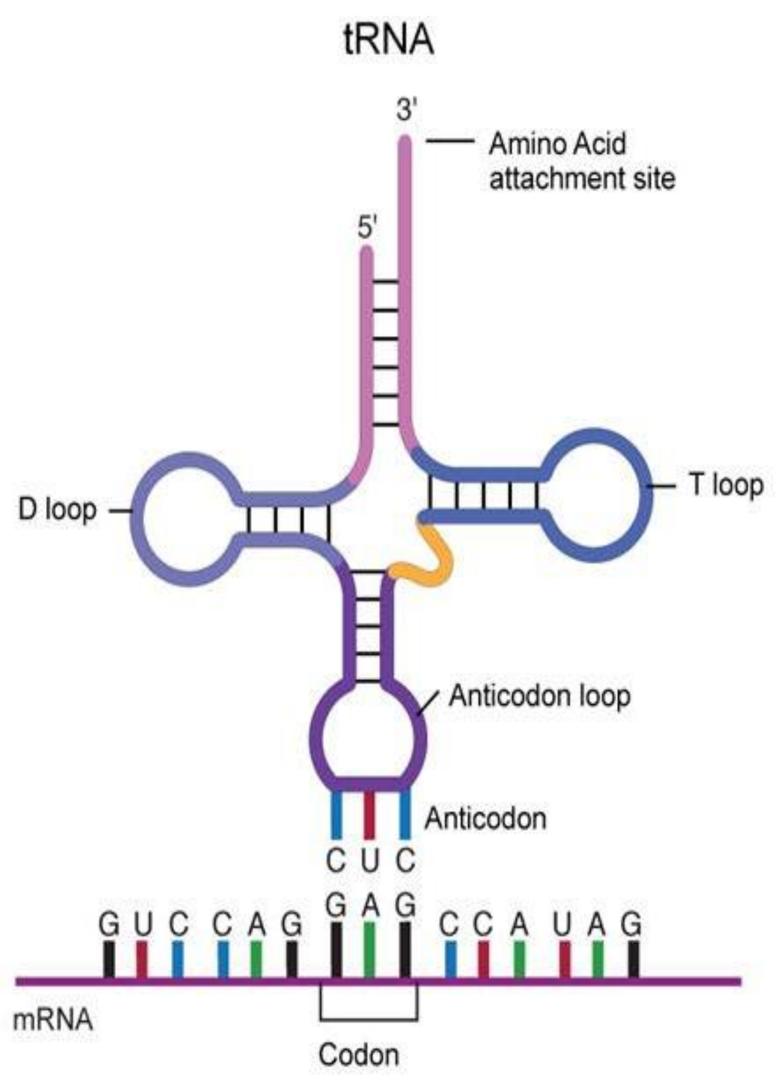
b- Transfer RNA (tRNA):

- It is a small RNA chain of that transfers a specific amino acid to a growing polypeptide chain at the ribosomal site of protein synthesis during translation. Vary in length from 74 to 95 nucleotides
- The tRNA molecules Serve as adapters; it has sites for amino acid attachment and an anti-codon region for codon recognition that binds to a specific sequence on the messenger RNA.



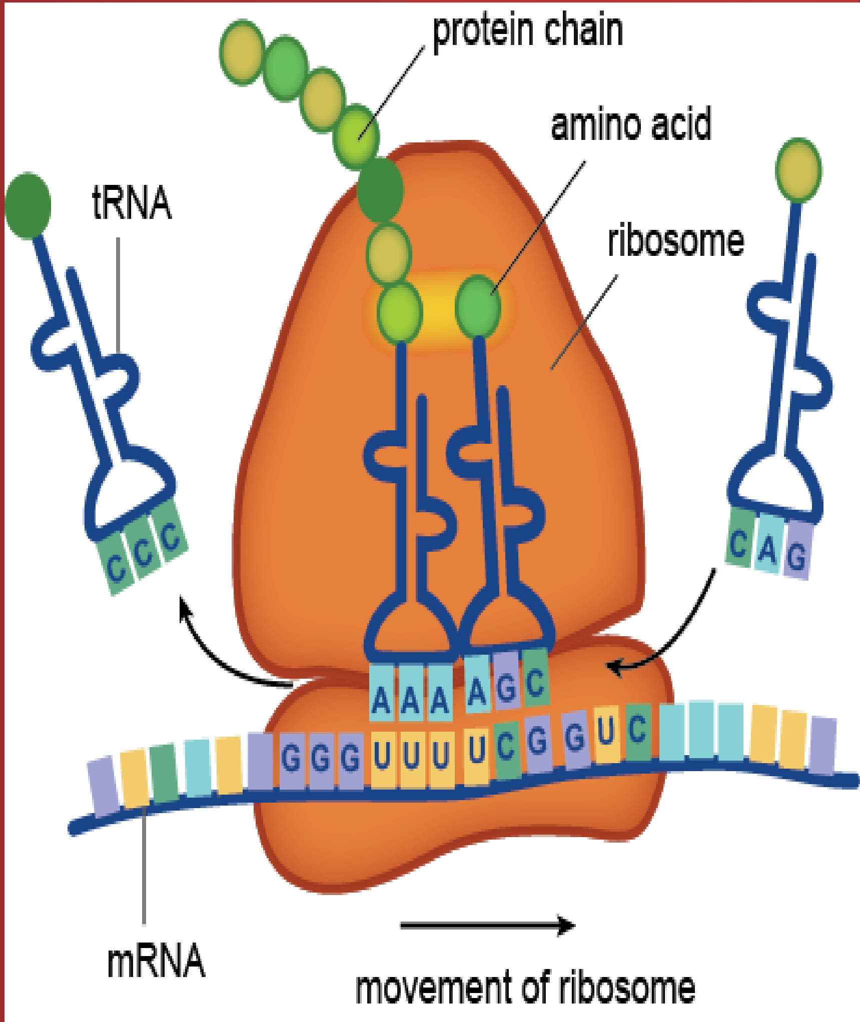
b- Transfer RNA (tRNA):

- There are at least 20 species of tRNA molecules in every cell, at least one (and often several) corresponding to each of the 20 amino acids required for protein synthesis.
- The secondary structure appears like a cloverleaf.
- tRNAs are stable in prokaryotes and they are somewhat less stable in eukaryotes.
- mRNAs, unstable in prokaryotes but generally stable in eukaryotic organisms.



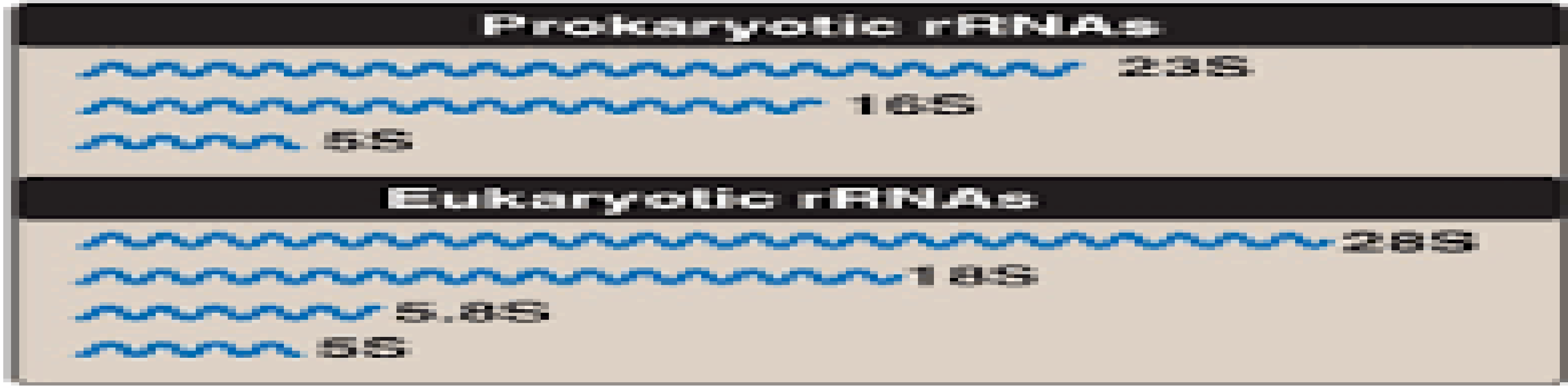
c- Ribosomal RNA (rRNA):

- A ribosome is a cytoplasmic nucleoprotein structure that acts as the machinery for the synthesis of proteins from the mRNA templates.
- On the ribosomes, the mRNA and tRNA molecules interact to translate into specific protein molecule information transcribed from the gene.

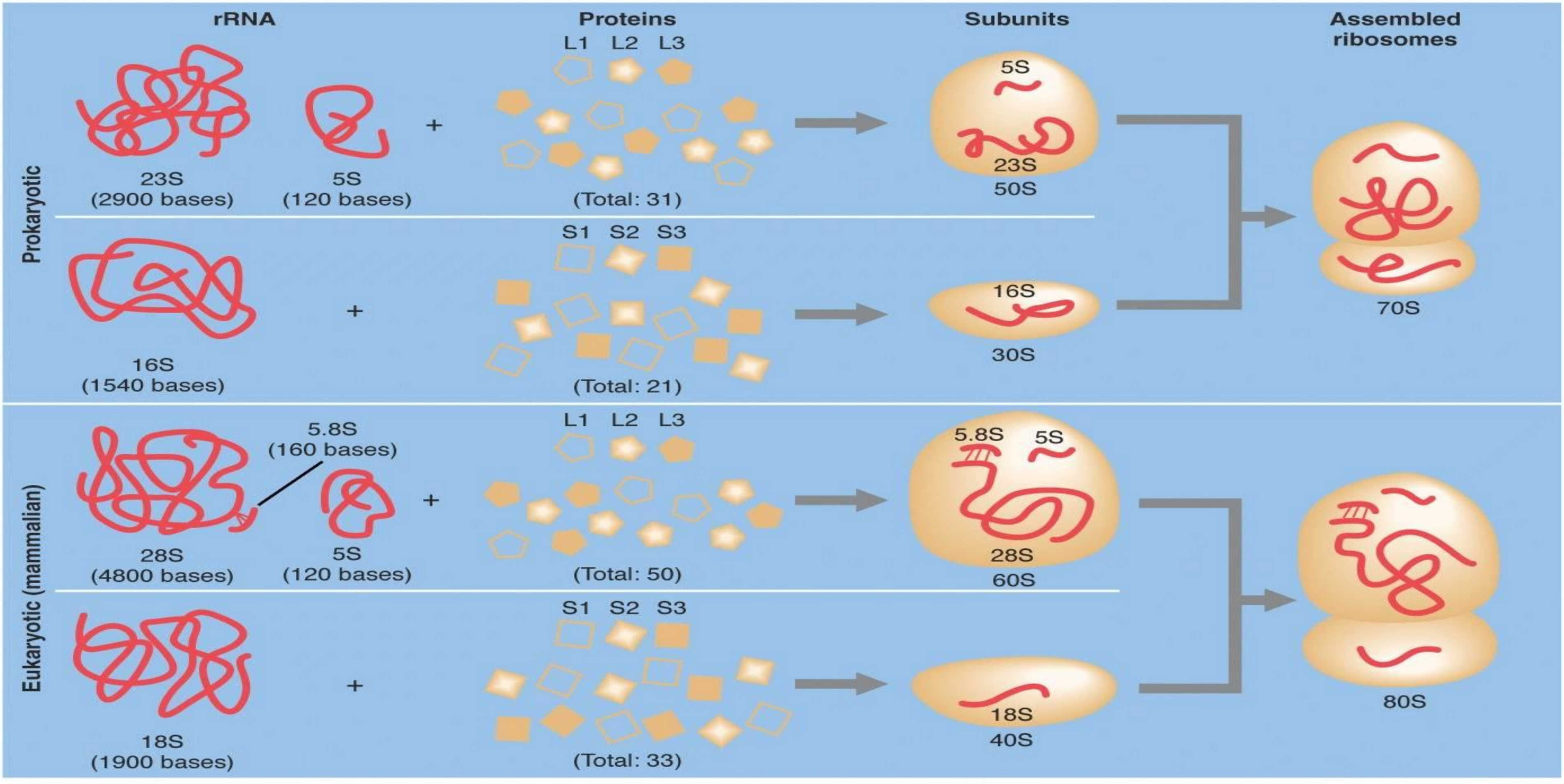


c- Ribosomal RNA (rRNA):

- The components ribosome were initially characterized by how rapidly they would "sink" in a centrifuge tube-in other words, they were described by their sedimentation velocity as measured in Svedberg (S) units.
- Both the eukaryotic and prokaryotic genomes contain multiple copies of these rRNA genes to be able to manufacture the large number of ribosome's required by a cell.
- The ribosome consists of two different subunits, one small and one large and is built up from rRNA and proteins.
- Inside the ribosome the amino acids are linked together into a chain through multiple biochemical reactions. Several ribosomes may be attached to a single mRNA at any time.



Prokaryotic rRNAs
23S
16S
5S
Eukaryotic rRNAs
28S
18S
5.8S
5S



rRNA
Proteins
Subunits
Assembled ribosomes
Prokaryotic
L1 L2 L3
23S (2900 bases)
5S (120 bases)
(Total: 31)
5S
23S
50S
S1 S2 S3
16S (1540 bases)
(Total: 21)
16S
30S
70S
Eukaryotic (mammalian)
L1 L2 L3
5.8S (160 bases)
28S (4800 bases)
5S (120 bases)
(Total: 50)
5.8S
5S
28S
60S
S1 S2 S3
18S (1900 bases)
(Total: 33)
18S
40S
80S

d- Viral RNA genomes:

- Like DNA, RNA can carry genetic information. RNA viruses have genomes composed of RNA which encodes a number of proteins. Viruses replicate their genomes by reverse transcribing DNA copies from their RNA; these DNA copies are then transcribed to new RNA.

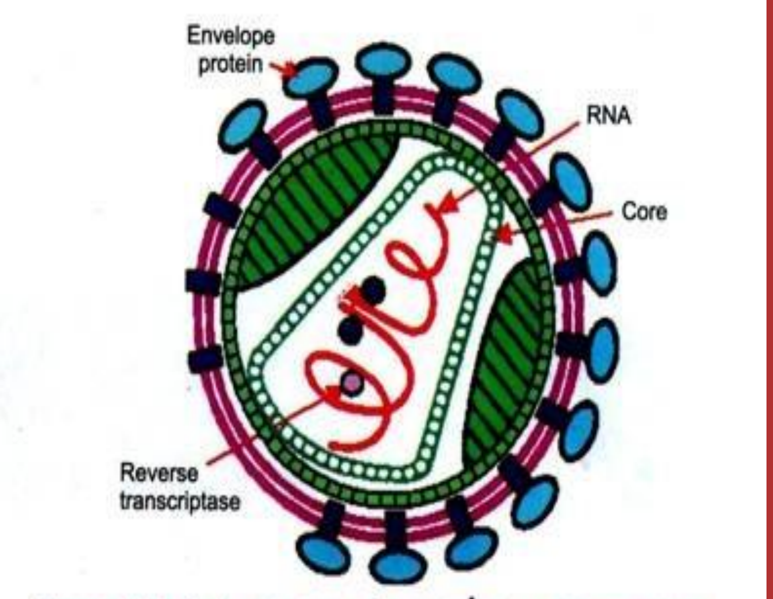
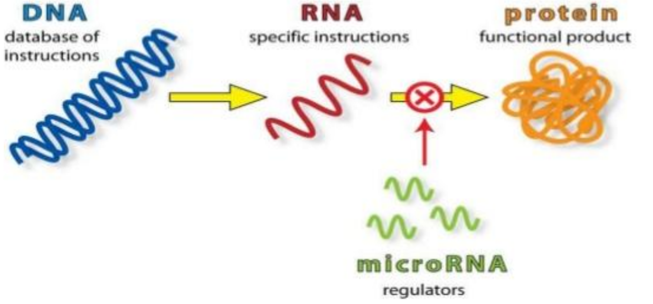


Image 11.4. Basic structure of a typical virus (diagrammatic).

E- Regulatory RNAs:

- In addition to mRNA, rRNA and tRNA, which play central roles within cells, there are a number of regulatory, non-coding RNAs (ncRNAs). Their main function is posttranscriptional regulation of gene expression.
- Many ncRNAs have been identified and characterized both in prokaryotes and eukaryotes, and are involved in the specific recognition of cellular nucleic acid targets through complementary base pairing, controlling cell growth and differentiation.



microRNAs, short non-coding RNAs present in all living organisms, have been shown to regulate the expression of at least half of all human genes. These single-stranded RNAs exert their regulatory action by binding messenger RNAs and preventing their translation into proteins.

Non-coding RNA	Length (nt)	Species	Function
Ribosomal RNA (rRNA)	120~4700	All	Translation
Transfer RNA (tRNA)	70~100	All	Translation
Small nuclear RNA (snRNA)	70~350	Eukaryote	Splicing, mRNA processing
Small nucleolar RNA (snoRNA)	70~300	Eukaryote, archaea	**RNA modification, rRNA processing**
miRNA	21~25	Eukaryote	Translational regulation
siRNA Small ncRNA	21~25	Eukaryote	Protection against viral infection
piRNA	24~30	Eukaryote	Genome stabilization
Long ncRNA	several hundreds~ several hundred thousands	Eukaryote	Transcription, splicing, transport regulation